

STAT452

Introduction to Design of Experiment: ANOVA

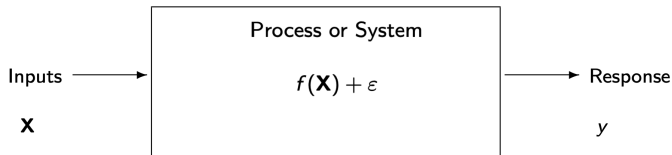
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Table of Content

- 1 Introduction to Design of Experiments
- 2 One-way Analysis of Variance (ANOVA)
- 3 Checking Model Assumptions
- 4 Randomized Complete Block Design (RCBD)

Introduction to Design of Experiments

- Design of Experiment – DOE is a systematic and efficient statistical method that allows scientists and engineers to investigate the relationship between multiple input variables (also called factors) and the output variables of interest (responses).



- Questions of interest for the above process: How does the variation in $\mathbf{X}$ affect the mean response? Which levels of the input variables will maximize the mean response? How should the input variables be combined to yield results with the smallest uncertainty?

Introduction to Design of Experiments

- Design of experiments is very useful in many scientific and engineering fields to address important problems: discovering new fundamental phenomena that may lead to the development of new products, new manufacturing processes, or improvements to existing processes and products, ...
- Most processes can be described in terms of a number of controllable variables such as time, temperature, concentration, ... By using experimental designs, engineers can determine which subset of process variables has the most significant impact on process performance and quality. The results of such experiments can help:
 - improve process productivity,
 - reduce process variability and achieve target performance,
 - shorten design and development time,
 - reduce operating costs.

Introduction to Design of Experiments

Some applications of statistical design of experiments in [engineering design](#) include:

- evaluating and comparing basic design configurations,
- assessing different materials,
- selecting design parameters so that the product performs well under various conditions,
- identifying key design parameters that affect product performance/quality.

Some Key Elements of Experimental Design

- **Randomization:** a crucial component of any experiment that helps avoid drawing biased or misleading conclusions.
- **Replication:** reduces **uncertainty** and increases precision by minimizing experimental variability.
- **Blocking:** reduces uncertainty by eliminating the effects of certain **nuisance factors**.
- **Factorial designs:** analyzing multiple factors simultaneously is more efficient than one-factor-at-a-time, and allows investigation of interactions among different factors.
- **Confounding:** occurs when the effects of two factors on a response variable cannot be distinguished (e.g., see **Simpson's paradox**). This can be avoided through proper experimental design.

Confounding

Example 1 (Simpson's paradox)

Recovery rates of 700 patients who were given a treatment drug were recorded, with 350 patients choosing to take the drug and 350 not taking it. The study results are shown in the following table:

	Drug	No drug
Men	81 out of 87 recovered (93%)	234 out of 270 recovered (87%)
Women	192 out of 263 recovered (73%)	55 out of 80 recovered (69%)
Combined data	273 out of 350 recovered (78%)	289 out of 350 recovered (83%)

Question: Should the doctor prescribe the drug to the patients?

Steps for Planning, Conducting, and Analyzing an Experiment

- 1 Identify and state the problem
- 2 Choose factors, levels, and ranges
- 3 Select the response variable
- 4 Determine the design
- 5 Conduct the experiment
- 6 Perform statistical analysis
- 7 Draw conclusions and make recommendations

One-way ANOVA: Example

Example 2 (Tensile strength / cf. Montgomery, 13-2)

A manufacturer of paper used for making grocery bags is interested in improving the product's tensile strength. Product engineering believes that tensile strength is a function of the hardwood concentration in the pulp and that the range of hardwood concentrations of practical interest is between 5 and 20%. A team of engineers responsible for the study decides to investigate four levels of hardwood concentration: 5%, 10%, 15%, and 20%. They decide to make up six test specimens at each concentration level by using a pilot plant. All 24 specimens are tested on a laboratory tensile tester in random order. The data from this experiment are shown in the table below.

Hardwood Concentration (%)	Observations						Totals	Averages
	1	2	3	4	5	6		
5	7	8	15	11	9	10	60	10.00
10	12	17	13	18	19	15	94	15.67
15	14	18	19	17	16	18	102	17.00
20	19	25	22	23	18	20	127	21.17
							383	15.96

Table 1: Tensile Strength of Paper (psi)

One-way Analysis of Variance

- **Question:** Is there a difference in tensile strength (psi) between packaging materials with four different levels of hardwood concentration in the pulp?
- The example addresses the problem of comparing means across multiple groups (≥ 3).
- The experimental design in this example is called a **Completely Randomized Design (CRD)**:
 - each **experimental unit** has an equal chance of receiving any of the **treatments**,
 - the assignment of treatments to experimental units is done independently.

Model for One-Factor Experiment

- Suppose we are interested in comparing k levels of a single factor.
- N experimental units are randomly selected to receive one of k treatments, where n_i units are assigned to treatment i , for $i = 1, \dots, k$.
- The observed data are displayed as in Table 1, where each value is denoted y_{ij} and represents the j -th observation under treatment i .
- The data in Table 1 can be modeled using the following linear model:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \quad (1)$$

for $i = 1, 2, \dots, k$ and $j = 1, 2, \dots, n_i$. Where:

- y_{ij} is the j -th observation in group i , and μ is the overall mean,
- τ_i is the effect of treatment i ,
- ε_{ij} is the random error term.

Model for One-Factor Experiment

- We can rewrite model (1) as:

$$y_{ij} = \mu_i + \varepsilon_{ij}, \quad (2)$$

where $\mu_i = \mu + \tau_i$ is the mean of treatment i .

- Assumptions about the error term: the errors are assumed to be independent, normally distributed with mean zero, and constant variance σ^2 .
- For a **fixed-effects model**, we assume that the sum of the treatment effects is zero, that is:

$$\sum_{i=1}^k \tau_i = 0. \quad (3)$$

Model for One-Factor Experiment

The data under model (2) can be described in the following table:

Treatment	Observations				Totals	Averages
1	y_{11}	y_{12}	$\cdots$	y_{1n_1}	$y_{1\cdot}$	$\bar{y}_{1\cdot}$
2	y_{21}	y_{22}	$\cdots$	y_{2n_2}	$y_{2\cdot}$	$\bar{y}_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	y_{k1}	y_{k2}	$\cdots$	y_{kn_k}	$y_{k\cdot}$	$\bar{y}_{k\cdot}$
					$y_{\cdot\cdot}$	$\bar{y}_{\cdot\cdot}$

where:

$$y_{i\cdot} = \sum_{j=1}^{n_i} y_{ij}, \quad \bar{y}_{i\cdot} = y_{i\cdot}/n_i, \quad i = 1, 2, \dots, k,$$

$$y_{\cdot\cdot} = \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}, \quad \bar{y}_{\cdot\cdot} = y_{\cdot\cdot}/N, \quad N = \sum_{i=1}^k n_i.$$

One-way ANOVA: Null and Alternative Hypotheses

- Null hypothesis:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k,$$

the means of all treatments are equal, i.e., there is no difference in group means.

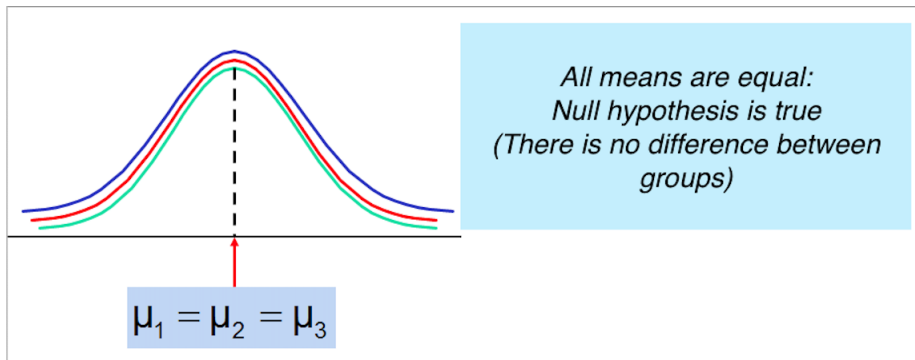
- Alternative hypothesis:

$$H_1 : \mu_i \neq \mu_j \quad \text{for at least one pair } 1 \leq i, j \leq k,$$

meaning there exists a difference between groups.

One-way ANOVA: Hypotheses

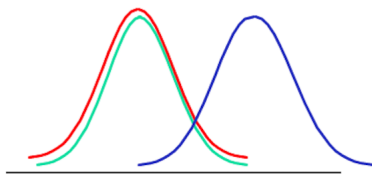
$$\begin{cases} H_0 : \mu_1 = \mu_2 = \dots = \mu_k, \\ H_1 : \mu_i \neq \mu_j \text{ for at least one pair } 1 \leq i, j \leq k \end{cases}$$



Hypotheses

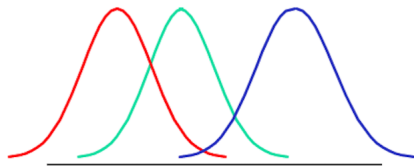
$$\begin{cases} H_0 : \mu_1 = \mu_2 = \dots = \mu_k, \\ H_1 : \mu_i \neq \mu_j \text{ for at least one pair } 1 \leq i, j \leq k \end{cases}$$

*Null hypothesis is false:
There is a difference between groups*



$$\mu_1 = \mu_2 \neq \mu_3$$

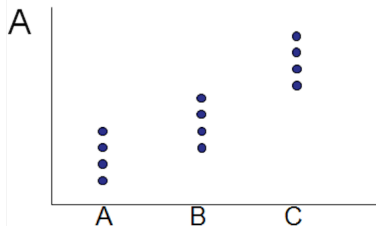
OR



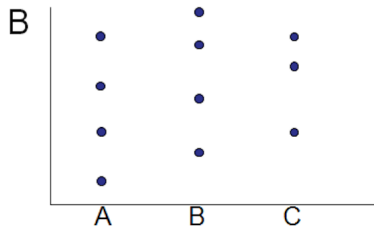
$$\mu_1 \neq \mu_2 \neq \mu_3$$

Analysis of Variation in the Data

- In ANOVA, the variation in the data is key to testing the equality of means across groups.



Small variation within groups



Large variation within groups

Sum of Squares

The total variation in the data can be decomposed into two components as follows:

$$SST = SSB + SSE, \quad (4)$$

$$\sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2. \quad (5)$$

Where:

- SST = Total Sum of Squares.
- SSB = Sum of Squares Between Groups.
- SSE = Error Sum of Squares (also denoted as SSW = Within-Groups Sum of Squares).

In the case of equal sample sizes for each group, $n_1 = n_2 = \dots = n_k = n$:

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2. \quad (6)$$

Mean Squares

Mean squares are calculated by dividing the corresponding sum of squares by their degrees of freedom:

- Total mean square:

$$MST = \frac{SST}{N - 1}, \quad (7)$$

where $N = \sum_{i=1}^k n_i$ (or $N = kn$ if all groups have equal size).

- Mean square between groups:

$$MSB = \frac{SSB}{k - 1}. \quad (8)$$

- Mean square error:

$$MSE = \frac{SSE}{N - k}. \quad (9)$$

Estimation of Variance – *MSE*

- When we have only one sample $y_1, y_2, \dots, y_n$, the population variance σ^2 is estimated by the sample variance:

$$s^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}.$$

- For the model $y_{ij} = \mu_i + \varepsilon_{ij}$, since all groups $i = 1, \dots, k$ have the same variance σ^2 , the sample variance s_i^2 in any group i can be used to estimate the common variance σ^2 , where:

$$s_i^2 = \frac{\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2}{n_i - 1}, \quad i = 1, 2, \dots, k.$$

Estimation of Variance – *MSE*

- We can pool all $s_1^2, s_2^2, \dots, s_k^2$ to obtain a better estimator for σ^2 , given by:

$$\begin{aligned}\hat{\sigma}^2 &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \dots + (n_k - 1)s_k^2}{(n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1)} \\ &= \frac{\sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2}{N - k} = \frac{SSE}{N - k} = MSE.\end{aligned}$$

- We can easily verify that *MSE* is an unbiased estimator of σ^2 because, within each group, s_i^2 is an unbiased estimator of σ^2 , so:

$$\begin{aligned}\mathbb{E}(SSE) &= \mathbb{E} \left(\sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 \right) = \mathbb{E} \left(\sum_{i=1}^k (n_i - 1)s_i^2 \right) \\ &= \sum_{i=1}^k (n_i - 1)\sigma^2 = (N - k)\sigma^2.\end{aligned}$$

Hence, $\mathbb{E}(MSE) = \mathbb{E}(SSE/(N - k)) = \sigma^2$.

One-way ANOVA Test

$$\begin{cases} H_0 : \mu_1 = \mu_2 = \dots = \mu_k, \\ H_1 : \mu_i \neq \mu_j \quad \text{for at least one pair } 1 \leq i, j \leq k \end{cases}$$

- Under $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$, we expect that

$$\bar{y}_{1.} \approx \bar{y}_{2.} \approx \dots \approx \bar{y}_{k.},$$

meaning that $\bar{y}_{i.} \approx \bar{y}_{..}$ for all treatments i . Therefore, a large value of $SSB = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{y}_{i.} - \bar{y}_{..})^2$ provides evidence against H_0 .

- On the other hand, if σ^2 is large $\Rightarrow$ large variation within groups even when H_0 is true. Thus, we should consider SSB/σ^2 instead of just SSB .
- σ^2 is estimated by MSE .

F-test Statistic

$$\begin{cases} H_0 : \mu_1 = \mu_2 = \dots = \mu_k, \\ H_1 : \mu_i \neq \mu_j \text{ for at least one pair } 1 \leq i, j \leq k \end{cases}$$

- The proposed test statistic is:

$$F_0 = \frac{MSB}{MSE} = \frac{SSB/(k-1)}{SSE/(N-k)} = \frac{\text{Between-group variation}}{\text{Within-group variation}}. \quad (10)$$

F_0 is the ratio of the estimated variance between groups to the estimated variance within groups.

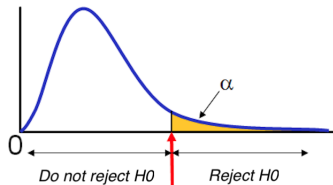
- The statistic F_0 follows an F distribution with degrees of freedom $df_1 = k - 1$ and $df_2 = N - k$.

One-way ANOVA Table

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between Groups	<i>SSB</i>	$k - 1$	<i>MSB</i>	$F = \frac{MSB}{MSE}$
Within Groups	<i>SSE</i>	$N - k$	<i>MSE</i>	
Total	<i>SST</i>	$N - 1$		

- Reject H_0 when:

$$F_0 > F_{\alpha; k-1, N-k} \quad (11)$$



One-way ANOVA: Example

EXAMPLE 1. (continued)

With $\alpha = 0.01$, we test the hypotheses:

$$\begin{cases} H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 \\ H_1 : \mu_i \neq \mu_j \text{ for at least one pair } 1 \leq i, j \leq 4 \end{cases}$$

Compute the relevant sums of squares. From Table 1, we have:

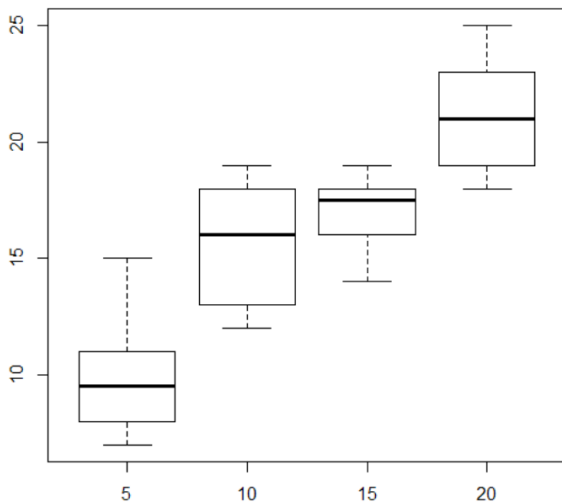
$$SST = \sum_{i=1}^4 \sum_{j=1}^6 y_{ij}^2 - \frac{y_{..}^2}{N} = (7^2 + 8^2 + \dots + 20^2) - \frac{383^2}{24} = 512.96,$$

$$SSB = \sum_{i=1}^4 \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N} = \frac{60^2 + 94^2 + 102^2 + 127^2}{6} - \frac{383^2}{24} = 382.79,$$

$$SSE = SST - SSB = 512.96 - 382.79 = 130.17.$$

One-way ANOVA: Example

Boxplot:



One-way ANOVA: Example

- Compute the test statistic:

$$F_0 = \frac{MSB}{MSE} = \frac{SSB/(k-1)}{SSE/(N-k)} = \frac{382.79/3}{130.17/20} = 19.60.$$

Since $F_{0.01;3,20} = 4.94$, we reject H_0 and conclude that hardwood concentration in the material affects the tensile strength of the paper bags.

- Conclusion using the p -value:

$$p = \mathbb{P}(F_{3,20} > 19.60) = 3.59 \times 10^{-6},$$

Since $p = 3.59 \times 10^{-6} \ll 0.01$, we have strong evidence to reject H_0 .

ANOVA with Unequal Sample Sizes

- When the sample sizes across treatments are not equal, the formulas for sum of squares need to be adjusted. Consider the ANOVA problem with k treatments, where the i -th treatment ($i = 1, 2, \dots, k$) has a sample size of n_i . The total number of observations is $N = \sum_{i=1}^k n_i$. The adjusted formulas for SST , SSB , and SSW are as follows:

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{..}^2}{N} \quad (12)$$

$$SSB = \sum_{i=1}^k \frac{y_{i.}^2}{n_i} - \frac{y_{..}^2}{N} \quad (13)$$

$$SSW = SST - SSB \quad (14)$$

Multiple Comparisons after ANOVA

- When the null hypothesis $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ is rejected in ANOVA, we know that there are differences among the groups (treatments), but we do not know specifically which groups differ.
- To identify which groups are significantly different, we use **multiple comparison methods**. A simple method is Fisher's **Least Significant Difference (LSD)** procedure.
- The LSD method compares all pairs of means using the hypothesis $H_0 : \mu_i = \mu_j$ (for all $i \neq j$), with the test statistic:

$$T = \frac{\bar{y}_{i\cdot} - \bar{y}_{j\cdot}}{\sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}. \quad (15)$$

Multiple Comparisons after ANOVA

- For a two-sided alternative, the means μ_i and μ_j are said to be significantly different if:

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > LSD$$

where

$$LSD = t_{\alpha/2}^{N-k} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \quad (16)$$

Here, N is the total number of experimental units, k is the number of treatments, and n_i , n_j are the sample sizes for treatments i and j , respectively.

Multiple Comparisons after ANOVA: Example

Applying multiple comparisons to the ANOVA results in Example 1, we have $k = 4$, $n = 6$, $MSE = 6.51$, and $t_{0.975}^{20} = 2.086$. The sample means for the four groups are:

$$\bar{y}_{1\cdot} = 10.00, \quad \bar{y}_{2\cdot} = 15.67, \quad \bar{y}_{3\cdot} = 17.00, \quad \bar{y}_{4\cdot} = 21.17.$$

The LSD value is $t_{0.975}^{20} \sqrt{2MSE/n} = 2.086 \sqrt{2 \times 6.51/6} = 3.07$. Pairwise comparisons of means:

$$4 \text{ vs. } 1 = 21.17 - 10.00 = 11.17 > 3.07,$$

$$4 \text{ vs. } 2 = 21.17 - 15.67 = 5.50 > 3.07,$$

$$4 \text{ vs. } 3 = 21.17 - 17.00 = 4.17 > 3.07,$$

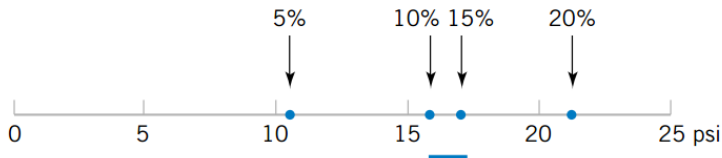
$$3 \text{ vs. } 1 = 17.00 - 10.00 = 7.00 > 3.07,$$

$$3 \text{ vs. } 2 = 17.00 - 15.67 = 1.33 < 3.07,$$

$$2 \text{ vs. } 1 = 15.67 - 10.00 = 5.67 > 3.07.$$

Multiple Comparisons after ANOVA: Example

From the analysis above, we see that except for groups 2 and 3, all other group means differ significantly. This indicates that hardwood concentrations of 10% and 15% in the pulp yield products with similar tensile strength. The result can be illustrated in the following plot:



Checking Model Assumptions

- Recall the assumptions about the model's error terms:
 - 1 Errors are independent of each other
 - 2 Errors follow a normal distribution
 - 3 Errors have constant variance
- Techniques for checking these assumptions often rely on the **predicted values** $\hat{y}_{ij}$ and the **residuals** $\hat{\epsilon}_{ij}$ (which are estimates of the unobserved error terms).

Residual Plot Analysis

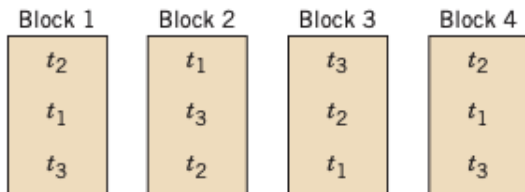
- ➊ Checking normality:
 - Draw a histogram of the residuals.
 - Create a normal QQ-plot of the residuals.
 - Use the Shapiro–Wilk or Kolmogorov–Smirnov test on the residuals.
- ➋ Checking constant variance:
 - Create a residual plot: plot $\hat{\epsilon}_{ij}$ against $\hat{y}_{ij}$.
 - Use Levene's test.
- ➌ Checking independence:
 - Plot the residuals $\hat{\epsilon}_{ij}$ against time/spatial variables.
 - Use the Durbin–Watson test.
 - Plot residuals against other variables of interest and assess the pattern of the plot.
- ➍ Additionally, residual analysis can help detect outliers or unusual observations. These points often have large values of $|\hat{\epsilon}_{ij}|$.

Block Designs

- A **nuisance factor** is a factor that affects the response variable y but is not of primary interest to the experimenter.
- When designing an experiment, one should consider the possibility of nuisance factors:
 - If the nuisance factor is unknown, randomization helps avoid biased conclusions, but its influence may increase variance.
 - If the nuisance factor is known but cannot be controlled, we can use **Analysis of Covariance (ANCOVA)** to measure and remove its effect.
 - If the nuisance factor is known and controllable, we use **block designs**.

Block Designs

- A **block** is a group of experimental units that are similar in some way. Units within the same block tend to have similar response values when given the same treatment.
- **Block design**: Randomize treatments among the experimental units within each block.



Block Design: Example

Example 3 (Air Freshener Sale / cf. Dean & Voss, p. 427)

- ▶ Goal: comparing 4 price+display treatments on the sales of a brand of air fresheners.
 - ▶ Treatment A = high price + extra display
 - ▶ Treatment B = middle price + extra display
 - ▶ Treatment C = low price + extra display
 - ▶ Treatment D = middle price + no extra display
- ▶ conducted at 8 stores for 4 weeks and each treatment lasts for one week in each store
- ▶ Response: unit sales in a one-week period

Week	Store							
	1	2	3	4	5	6	7	8
1	B 31	A 23	C 12	D 3	A 10	C 30	B 23	D 14
2	A 19	D 16	B 14	C 4	B 21	D 25	C 17	A 14
3	D 15	C 30	A 12	B 6	C 12	A 47	D 5	B 3
4	C 16	B 27	D 5	A 11	D 12	B 38	A 13	C 6

Randomized Complete Block Design

- Consider b blocks, each containing a experimental units.
- Randomly assign the a treatments to the experimental units within each block.
- The RCBD data layout is represented as follows:

Treatments	Blocks				Totals	Averages
	1	2	...	b		
1	y_{11}	y_{12}	...	y_{1b}	$y_{1\cdot}$	$\bar{y}_{1\cdot}$
2	y_{21}	y_{22}	...	y_{2b}	$y_{2\cdot}$	$\bar{y}_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$
a	y_{a1}	y_{a2}	...	y_{ab}	$\bar{y}_{a\cdot}$	$\bar{y}_{a\cdot}$
Totals	$y_{\cdot 1}$	$y_{\cdot 2}$	...	$y_{\cdot b}$	$\bar{y}_{\cdot\cdot}$	
Averages	$\bar{y}_{\cdot 1}$	$\bar{y}_{\cdot 2}$	...	$\bar{y}_{\cdot b}$		$\bar{y}_{\cdot\cdot}$

- Identify variables to be used for **blocking** by locating sources of variation (e.g., age, gender, medical history, geographic location, ...), or by blocking over space, time, machine, etc.

Model for RCBD

- Use the linear model:

$$y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \quad \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \end{cases} \quad (17)$$

where:

- y_{ij} : response from the unit receiving treatment i in block j
- μ : overall mean
- α_i : effect of treatment i
- β_j : effect of block j
- ε_{ij} : error term, $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$
- Model constraints:

$$\sum_{i=1}^a \alpha_i = 0 \quad \text{and} \quad \sum_{j=1}^b \beta_j = 0. \quad (18)$$

Sum of Squares

- The observation y_{ij} can be expressed as:

$$\begin{aligned} y_{ij} &= \bar{y}_{..} + (\bar{y}_{i.} - \bar{y}_{..}) + (\bar{y}_{.j} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}) \\ &= \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + \hat{\epsilon}_{ij}. \end{aligned}$$

- Similar to the CRD, the total sum of squares $SST = \sum_{i=1}^a \sum_{j=1}^b (y_{ij} - \bar{y}_{..})^2$ is decomposed as:

$$SST = SSG + SSB + SSE,$$

where:

- $SSG = b \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2$: Sum of Squares for Treatments.
- $SSB = a \sum_{j=1}^b (\bar{y}_{.j} - \bar{y}_{..})^2$: Sum of Squares for Blocks.
- $SSE = \sum_{i=1}^a \sum_{j=1}^b (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$: Sum of Squares for Error.
- Degrees of freedom:

$$ab - 1 = (a - 1) + (b - 1) + (a - 1)(b - 1).$$

Mean Squares

- The mean squares in RCBD are given by:

$$MSG = \frac{SSG}{a-1}, \quad MSB = \frac{SSB}{b-1}, \quad MSE = \frac{SSE}{(a-1)(b-1)}.$$

- Based on the RCBD model $y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ and constraints $\sum_{i=1}^a \alpha_i = 0$ and $\sum_{j=1}^b \beta_j = 0$, we can show:

$$\mathbb{E}(MSG) = \sigma^2 + \frac{b}{a-1} \sum_{i=1}^a \alpha_i^2,$$

$$\mathbb{E}(MSB) = \sigma^2 + \frac{a}{b-1} \sum_{j=1}^b \beta_j^2,$$

$$\mathbb{E}(MSE) = \sigma^2.$$

Thus, MSE is still an unbiased estimator of σ^2 .

ANOVA Test for Treatment Effect

- For treatments (groups):

$$\begin{cases} H_{0a} : \alpha_1 = \alpha_2 = \dots = \alpha_a = 0 \\ H_{1a} : \text{Not all } \alpha_i \text{ are equal to 0.} \end{cases} \quad (19)$$

- Under H_{0a} , the test statistic is:

$$F_{\text{trt}} = \frac{MSG}{MSE} \sim F_{a-1, (a-1)(b-1)}. \quad (20)$$

- Reject H_{0a} if:

$$F_{\text{trt}} > F_{\alpha; a-1, (a-1)(b-1)}.$$

Two-Way ANOVA Table for RCBD

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Treatments (Groups)	<i>SSG</i>	$a - 1$	<i>MSG</i>	$F_{\text{trt}} = \frac{MSG}{MSE}$
Blocks	<i>SSB</i>	$b - 1$	<i>MSB</i>	$F_{\text{block}} = \frac{MSB}{MSE}$
Error	<i>SSE</i>	$(a - 1)(b - 1)$	<i>MSE</i>	
Total	<i>SST</i>	$ab - 1$		

The statistic F_{block} is used to test for block effects, but since block effects are usually not of primary interest, this is often ignored.

References

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